

TAMPINES JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION



CANDIDATE
NAME

CIVICS
GROUP

1	2			
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TUTOR
NAME

Biology

8875/02

Paper 2

Monday, 2 September 2013

2 hours

Candidates answer on the Question Paper

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

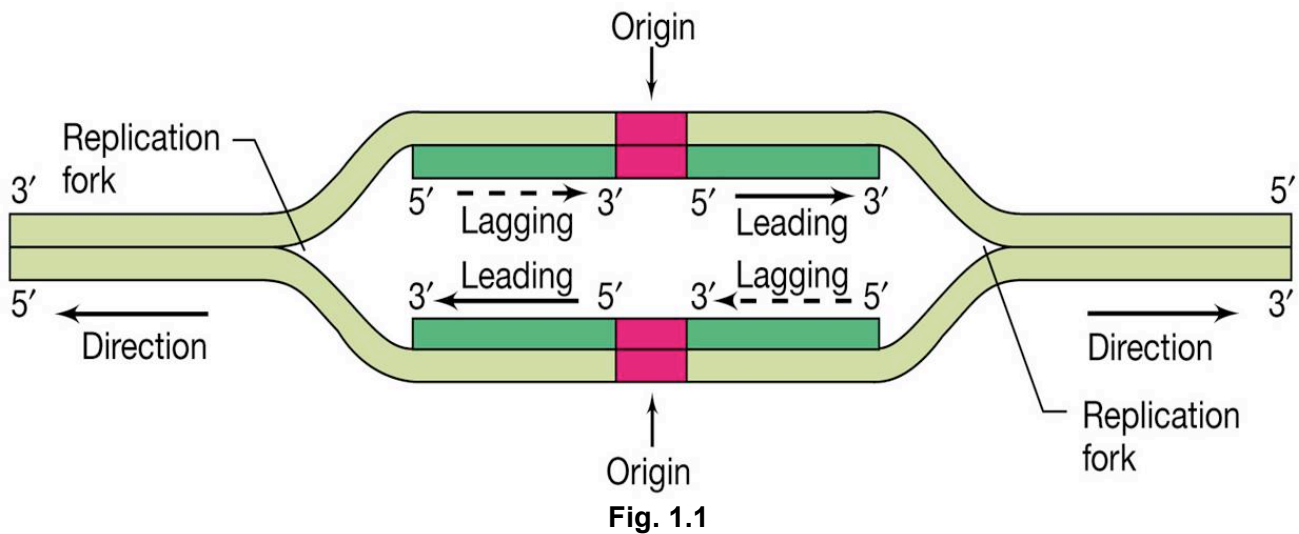
For Examiner's Use		
Section A		
1	10	
2	10	
3	10	
4	10	
Section B		
5 OR	20	
6	20	
Total	60	

SECTION A - STRUCTURED QUESTIONS (80 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION 1

Fig. 1.1 is a diagram showing part of a bacterial chromosome with a single origin of replication and two replication forks which proceed in opposite directions until replication of the entire bacterial chromosome is complete.



- (a) With reference to **Fig. 1.1**, explain why this replication is said to be semi-conservative. [2]

- (b) With reference to **Fig. 1.1**, replication only occurs in a 5' to 3' direction, explain why this is so. [2]

The Hayflick limit is the number of times a normal human cell population will divide until cell division stops. Empirical evidence shows that the telomeres, DNA regions that do not code for proteins, associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. The Hayflick limit has been found to correlate with the length of the telomere region at the end of a strand of DNA. **Fig. 1.2** shows the position of telomeres in a DNA double helix.

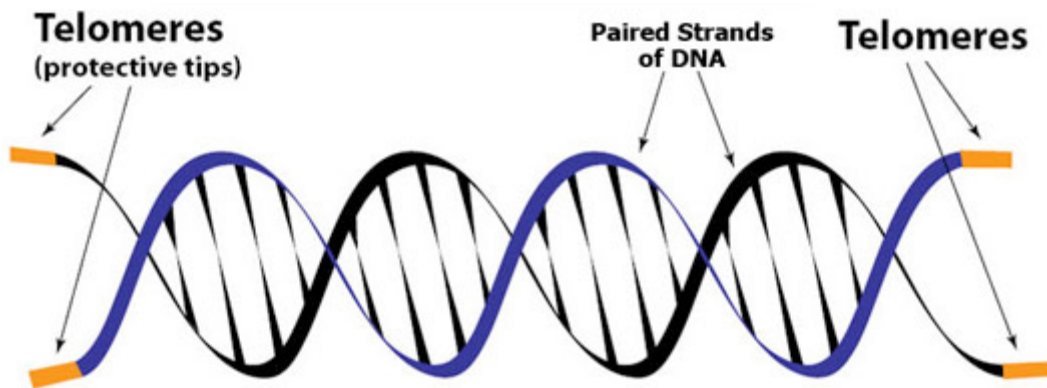


Fig. 1.2

- (c) Suggest the role of telomeres. [2]

In stem cells, the telomeres are protected from shortening with the help of telomerase. Telomerase is an enzyme that lengthens the telomeres. Having lengthened telomeres means that the cell can undergo extensive cellular division without shortening the telomeres to critical lengths.

- (d) Some stem cells, such as the zygote is said to be totipotent.

State what it is meant by Totipotent. [2]

- (e) Suggest how having the activity of Telomerase in stem cells allows them to divide indefinitely. [1]

[Total: 10]

QUESTION 2

Carbon is essential to life. All of our molecular machines are built around a central scaffolding of organic carbon. Unfortunately, carbon in the earth and atmosphere is locked in highly oxidized forms, such as carbonate minerals and carbon dioxide gas. In order to be useful, this oxidized carbon must be "fixed" into more organic forms, rich in carbon-carbon bonds and decorated with hydrogen atoms. Powered by the energy of sunlight, plants perform this central task of carbon fixation.

Inside plant cells, the enzyme ribulose biphosphate carboxylase (rubisco) forms the bridge between life and the lifeless, creating organic carbon from the inorganic carbon dioxide in the air. **Fig. 2.1** shows the structure of ribulose biphosphate carboxylase (rubisco).

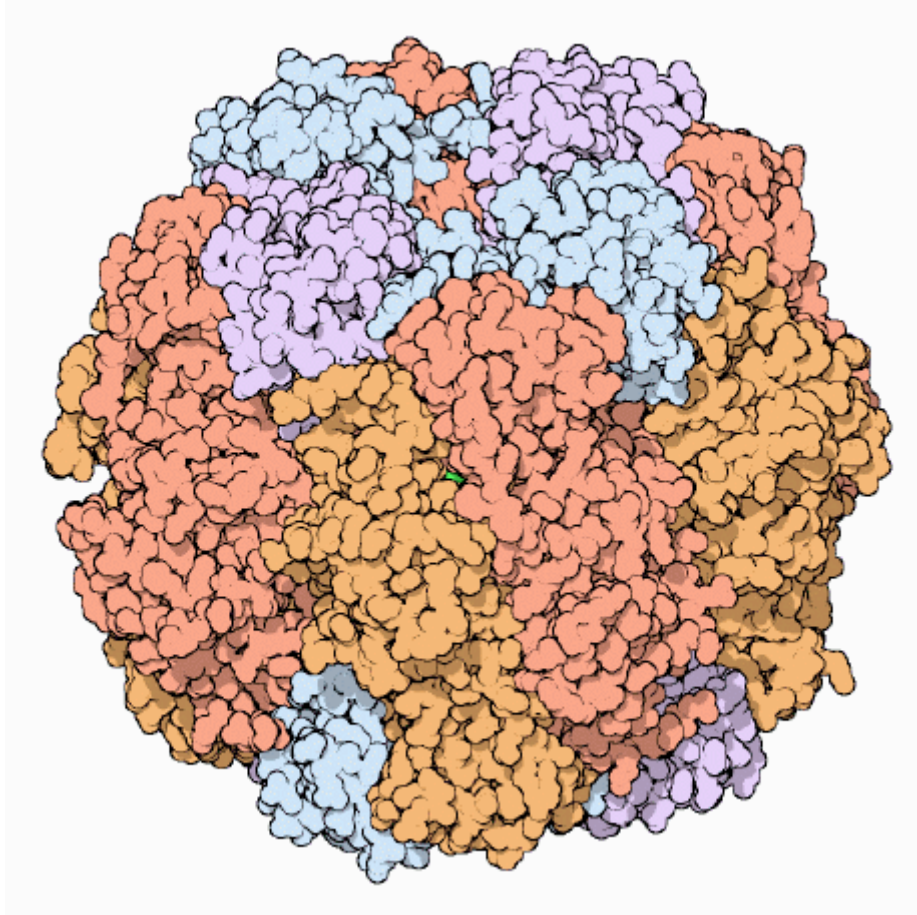


Fig. 2.1

- (a) Rubisco is an enzyme made of sixteen polypeptide chains.

Describe how the structure of rubisco is maintained. [2]

In spite of its central role, rubisco is remarkably inefficient. Typical enzymes can process a thousand molecules per second, but rubisco fixes only about three carbon dioxide molecules per second. **Fig. 2.2** shows the relationship between CO_2 assimilation rate and increasing light intensity in a plant, when carbon dioxide is present in excess.

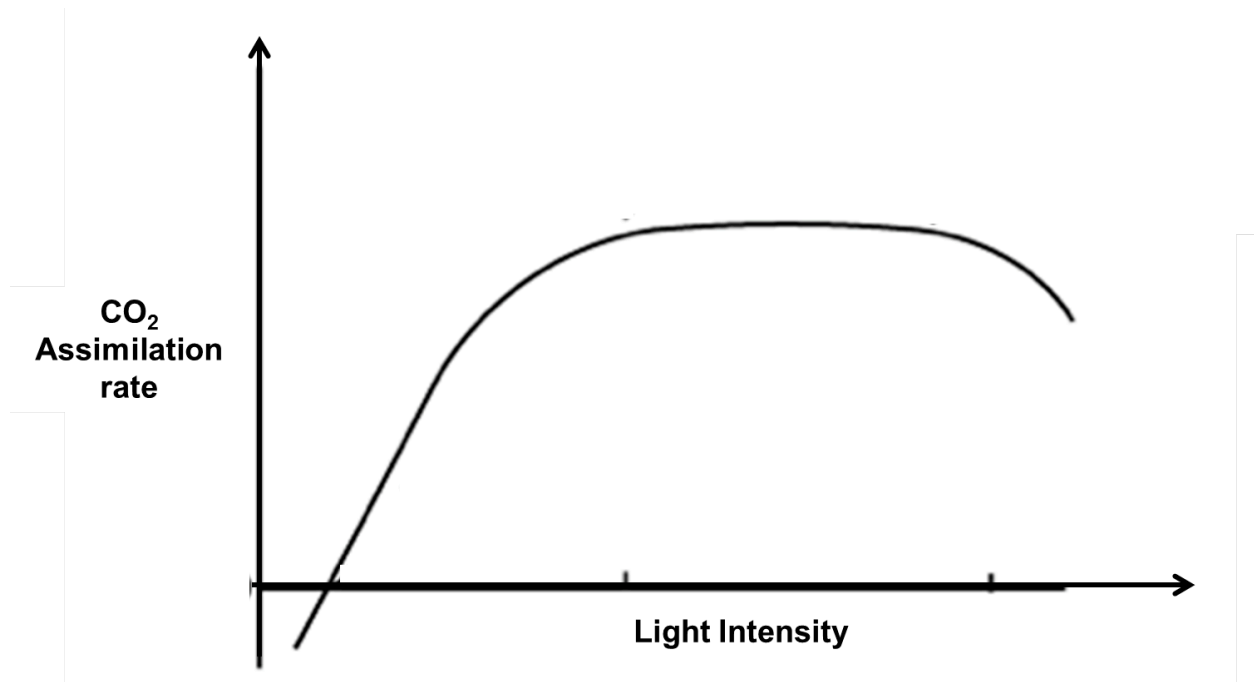


Fig. 2.2

- (b) Explain how light energy assists in the process of carbon fixation in plants. [3]

- (c) Explain why the CO_2 assimilation rate levels off at higher light intensity. [3]

Rubisco also shows an embarrassing lack of specificity. Unfortunately, oxygen molecules and carbon dioxide molecules are similar in shape and chemical properties. In rubisco, an oxygen molecule can bind comfortably in the active site designed to bind carbon dioxide. Rubisco then attaches the oxygen to the sugar chain, forming a faulty oxygenated product.

- (c) With reference to **Fig 2.2**, suggest why the CO_2 assimilation rate is declining as the light intensity increases higher. [2]

[Total: 10]

QUESTION 3

Fig. 3.1 shows Honeycreepers of the Family Drepanididae from the Hawaiian Islands. These Honeycreepers reside on the various Hawaiian Islands. Through a series of studies, these Honeycreepers are found to be evolved from an ancestral Honeycreeper which resided originally on, Kauai, one of the Hawaiian Islands.



Fig. 3.1

- (a) Explain how natural selection could have caused the evolution of these various species of Honeycreepers. [4]

Our understanding of the evolutionary relationship of species has been greatly improved through the use of molecular techniques.

Many evolutionary studies use mitochondrial DNA rather than nuclear DNA.

- (b) Explain why mitochondrial DNA is used to determine phylogeny instead of using nuclear DNA. [2]

A new Honeycreeper (**Fig. 3.2**) has been found on one of the Hawaiian Islands. Based on its structural features, Scientists are speculating that this new found Honeycreeper is also a descendant of the ancestral Honeycreeper found on Kauai.



Fig. 3.2

- (d) Suggest why it is not advisable to determine the evolutionary relationship of this new found Honeycreeper to the rest of them from the morphological features. [2]

- (c) Explain how you could use molecular homology to show that this new Honeycreeper evolved from the original ancestor Honeycreeper. [2]

[Total: 10]

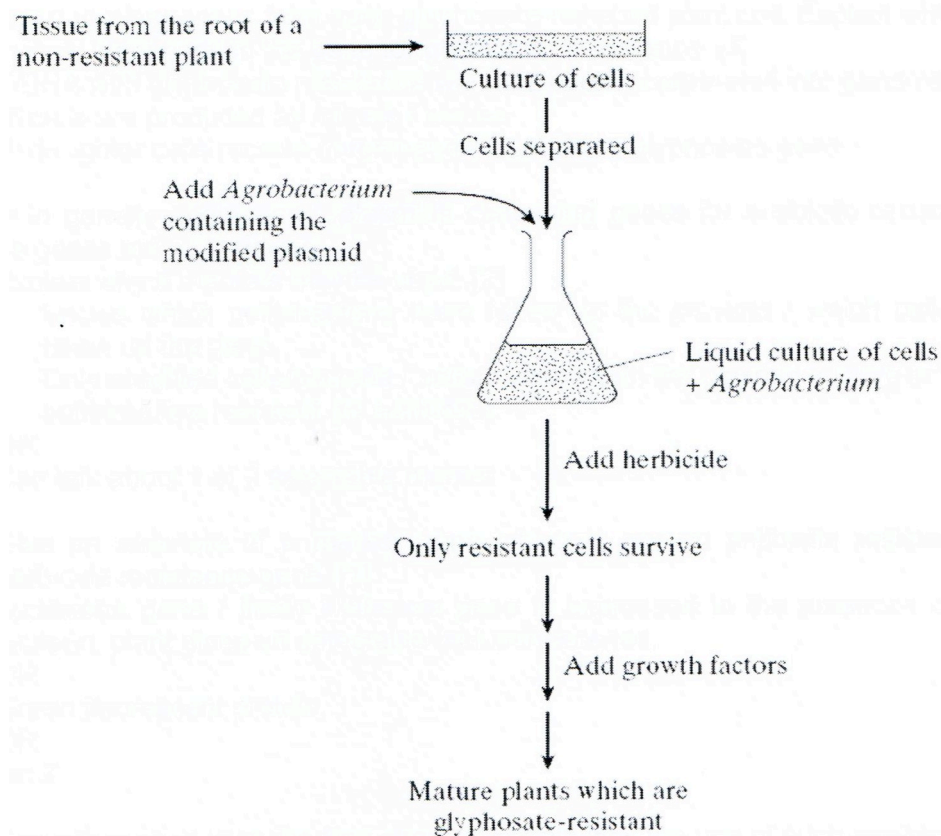
QUESTION 4

Read the following passage.

Genetic engineering is a technique for improving crop plants. It is a process by which a single new gene can be added to the genes already present in the plants. One way of doing this is to use a natural genetic engineer, a soil microorganism called *Agrobacterium*. This bacterium possesses a plasmid which can be modified in the laboratory so that it becomes the carrier of new genetic information.

For example, a gene coding for the ability to break down the herbicide glyphosate can be introduced with the assistance of the enzymes restriction endonuclease and ligase. Once inside the crop plant, the gene for glyphosate breakdown makes the plant resistant to the effects of this herbicide. All surrounding weeds are destroyed when sprayed with glyphosate.

The flow-chart shows how the plasmid carrying this gene can be used to produce glyphosate-resistant plants.



Use the information from the passage and your own knowledge to answer the following questions.

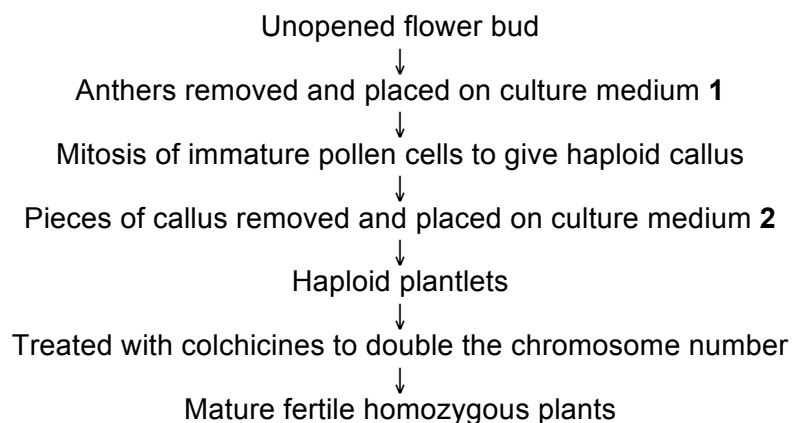
- (a) One mature plant grows from each glyphosate-resistant plant cell. Explain why all the cells of the mature plant contain the gene for glyphosate-resistance. [1]

- (b) Outline the production of modified plasmid containing the gene of interest. [3]

- (c) Suggest and explain **one** possible problem associated with the use of herbicides together with genetically modified, herbicide-resistant crop plants. [1]

Another technique required in plant biotechnology is plant tissue culture. Various organs or tissues of a plant may be used in tissue culture to produce a large number of genetically identical plants.

In the process of anther culture, haploid plantlets are produced from immature anthers, from which mature fertile homozygous plants are obtained. The process is shown diagrammatically in the figure below.



- (d) With reference to the figure,

- (i) Explain what is meant by haploid callus. [2]

- (ii) State one precaution that would be taken when transferring material onto culture

medium **1** or **2**. [1]

- (iii) Suggest why variation may be generated in plantlets derived from the same callus. [1]

- (iv) Suggest how colchicine could bring about the doubling of chromosome number in haploid cells. [1]

[Total: /10 marks]

SECTION B (20 MARKS)

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Answer **one** question.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION 5

- (a) Compare anaerobic respiration and aerobic respiration. [6]
- (b) Describe the main features of oxidative phosphorylation. [8]
- (c) Discuss how the mitochondrion is adapted for carrying out respiration. [6]

QUESTION 6

- (a) Describe PCR and compare and contrast with replication process. [8]
- (b) Outline the process of gel electrophoresis and explain how it is able to separate fragments of DNA. [9]
- (c) State the uses of RFLP analysis. [3]

End of Paper